

Preparatory Course for MSc Students, WiSe 2026/27

Exercise Sheets for Sections 1–4 of the lecture notes

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How to use these sheets. Each sheet corresponds to one section of the lectures. Exercises marked **(W)** are warm-ups meant to be done at the start of a session; unmarked exercises form the core of the session; exercises marked **(*)** are more challenging. Notation follows the lecture notes; $\mathbb{K}$ always denotes a field.

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Sheet 1: Algebraic Structures

Exercise 1.1 (W) *Semigroups, monoids, groups.* Decide in each case whether the given set with the given operation is a semigroup, a monoid, or a group, and whether it is commutative. Identify the neutral element and the invertible elements where they exist.

- (a) $(\mathbb{Z}, -)$.
- (b) $(\mathbb{N}, \max)$, where $\mathbb{N} = \{1, 2, 3, \dots\}$.
- (c) $(\mathbb{N}_0, \gcd)$, with the convention $\gcd(0, n) = n$.
- (d) $(\mathcal{P}(X), \triangle)$ for a set X , where $A \triangle B = (A \setminus B) \cup (B \setminus A)$ is the symmetric difference.
- (e) $(\mathbb{Z}, *)$ with $a * b = a + b + ab$.
- (f) $(\mathbb{R} \setminus \{-1\}, *)$ with the same operation $a * b = a + b + ab$.

Hint for (e), (f): compute $1 + (a * b)$.

Exercise 1.2 *Neutral elements and inverses.* Let $(M, *)$ be a monoid with neutral element e .

- (a) Show that the neutral element is unique.
- (b) Suppose $s \in M$ has a left inverse t ($t * s = e$) and a right inverse t' ($s * t' = e$). Show that $t = t'$. Conclude that inverses in a group are unique.
- (c) Show that $M^\times = \{s \in M \mid s \text{ invertible}\}$ is a group and that $(s * t)^{-1} = t^{-1} * s^{-1}$.
- (d) Show that $\text{Maps}(X, X)^\times = \text{Sym}(X)$.
- (e) Find an element of $\text{Maps}(\mathbb{N}, \mathbb{N})$ that has a left inverse but no right inverse. Why does this not contradict (b)?

Exercise 1.3 *Basic ring identities.* Let $(R, +, \cdot)$ be a ring.

- (a) Show that $0_R \cdot a = 0_R$ and $(-1_R) \cdot a = -a$ for all $a \in R$.
- (b) Show that if $1_R = 0_R$, then $R = \{0_R\}$.
- (c) Show that a unit is never a zero divisor.
- (d) Show that every *finite* integral domain is a field. *Hint:* for $a \neq 0$ consider the map $R \rightarrow R$, $x \mapsto ax$.
- (e) Deduce that $\mathbb{Z}/p\mathbb{Z}$ is a field for every prime p .

Exercise 1.4 *The rings $\mathbb{Z}/m\mathbb{Z}$.*

- (a) List all units and all zero divisors of $\mathbb{Z}/12\mathbb{Z}$.
- (b) Show that every non-zero element of $\mathbb{Z}/m\mathbb{Z}$ is either a unit or a zero divisor. *Hint:* distinguish $\gcd(a, m) = 1$ and $\gcd(a, m) > 1$; use Bézout.
- (c) Find the inverse of $[7]$ in $\mathbb{Z}/30\mathbb{Z}$ using the Euclidean algorithm.
- (d) Show that $\mathbb{Z}/m\mathbb{Z}$ is an integral domain if and only if m is prime.

Exercise 1.5 *Units in polynomial and matrix rings.*

- (a) Let R be an integral domain. Show that $\deg(fg) = \deg f + \deg g$ for non-zero $f, g \in R[x]$, and conclude that $R[x]^* = R^*$. In particular, $\mathbb{K}[x]^* = \mathbb{K} \setminus \{0\}$.

- (b) Show that $1 + 2x$ is a unit in $(\mathbb{Z}/4\mathbb{Z})[x]$. Which hypothesis in (a) fails?
- (c) Find a non-zero $A \in \text{Mat}(2 \times 2, \mathbb{R})$ with $A^2 = 0$, and show that $I + A$ is a unit.
- (d) Show that every non-zero $A \in \text{Mat}(2 \times 2, \mathbb{K})$ is either a unit or a (left and right) zero divisor.
Hint: consider $\text{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

Exercise 1.6 *Gaussian integers.* Let $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ and define $N(z) = z\bar{z} = a^2 + b^2$ for $z = a + bi$.

- (a) Verify that $\mathbb{Z}[i]$ is a subring of $\mathbb{C}$ and an integral domain.
- (b) Show that $N(zw) = N(z)N(w)$.
- (c) Determine the group of units $\mathbb{Z}[i]^*$. To which familiar group is it isomorphic?
- (d) Show that the localisation $S^{-1}\mathbb{Z}[i]$ with $S = \mathbb{Z}[i] \setminus \{0\}$ is isomorphic to $\mathbb{Q}(i) = \{p + qi \mid p, q \in \mathbb{Q}\}$.

Exercise 1.7 *Modules versus vector spaces.*

- (a) Let $(G, +)$ be an abelian group. Show that the module axioms force $n \cdot g = g + \cdots + g$ (n times) for $n \in \mathbb{N}$, $0 \cdot g = 0$ and $(-n) \cdot g = -(n \cdot g)$. Hence G carries *exactly one* $\mathbb{Z}$ -module structure.
- (b) Show that for $n \geq 2$ the abelian group $\mathbb{Z}/n\mathbb{Z}$ admits no structure of a $\mathbb{Q}$ -vector space (extending its addition).
- (c) Show that every $\mathbb{Z}$ -linear map $\mathbb{Q} \rightarrow \mathbb{Z}$ is zero.
- (d) Show that any two non-zero submodules of the $\mathbb{Z}$ -module $\mathbb{Z}$ intersect non-trivially. Conclude that $2\mathbb{Z} \subset \mathbb{Z}$ has no complementary submodule.
- (e) Let $T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ act on $V = \mathbb{R}^2$ and make V an $\mathbb{R}[x]$ -module via $p(x) \cdot v = p(T)v$. Show that $(x^2 + 1) \cdot v = 0$ for all v , and determine all $\mathbb{R}[x]$ -submodules of V .

Exercise 1.8 *The Grothendieck construction.* Let $(M, *)$ be a commutative monoid and $G(M) = (M \times M) / \sim$ as in the lecture.

- (a) Show that the operation $[(m, n)] * [(\tilde{m}, \tilde{n})] = [(m * \tilde{m}, n * \tilde{n})]$ is well defined. Show that $[(e, e)]$ is neutral and that $[(n, m)]$ is inverse to $[(m, n)]$.
- (b) Compute $G(\mathbb{N}, \cdot)$, where $\mathbb{N} = \{1, 2, \dots\}$.
- (c) Compute $G(\mathbb{N}_0, \cdot)$. *Hint:* the element t in the definition of $\sim$ may be 0.
- (d) Show that $\iota : M \rightarrow G(M)$, $m \mapsto [(m, e)]$, is a monoid homomorphism, and that it is injective if and only if M is *cancellative* ($m * t = m' * t \Rightarrow m = m'$).
- (e) (*) (Universal property.) Let H be an abelian group and $f : M \rightarrow H$ a monoid homomorphism. Show that there is a unique group homomorphism $F : G(M) \rightarrow H$ with $F \circ \iota = f$.

Exercise 1.9 *Localisation.* Let R be a commutative ring and $S \subset R$ multiplicative.

- (a) Show that $\frac{r}{1} = \frac{0}{1}$ in $S^{-1}R$ if and only if $tr = 0$ for some $t \in S$.
- (b) Show that $S^{-1}R = 0$ if $0 \in S$.
- (c) Let $S = \{1, 2, 4, 8, \dots\} \subset \mathbb{Z}$. Identify $S^{-1}\mathbb{Z}$ with a subring of $\mathbb{Q}$.

- (d) Let $R = \mathbb{Z}/6\mathbb{Z}$ and $S = \{[1], [2], [4]\}$. Check that S is multiplicative, determine the kernel of $R \rightarrow S^{-1}R$, $r \mapsto r/1$, and show that $S^{-1}R \cong \mathbb{Z}/3\mathbb{Z}$.
- (e) Explain why the factor t in the definition of $\sim$ is needed. *Hint:* in (d), compare the pairs $([0], [1])$, $([0], [2])$, $([3], [2])$ under the naive relation $(r, s) \approx (r', s') \iff rs' = r's$.

Exercise 1.10 *Subgroups and normal subgroups.*

- (a) Show that an intersection of subgroups is a subgroup. Is $2\mathbb{Z} \cup 3\mathbb{Z}$ a subgroup of $\mathbb{Z}$?
- (b) Show that every subgroup of index 2 is normal.
- (c) List all subgroups of S_3 and decide which of them are normal.
- (d) Show that $\mathrm{SL}(n, \mathbb{R})$ is normal in $\mathrm{GL}(n, \mathbb{R})$, but that $O(2)$ is *not* normal in $\mathrm{GL}(2, \mathbb{R})$. *Hint:* conjugate $\mathrm{diag}(1, -1)$ by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.
- (e) Show that every subgroup of $(\mathbb{Z}, +)$ is of the form $n\mathbb{Z}$.

Exercise 1.11 *Homomorphisms, kernels and ideals.*

- (a) Let $\varphi : G \rightarrow H$ be a group homomorphism. Show that $\varphi(e) = e$, $\varphi(g^{-1}) = \varphi(g)^{-1}$, and that φ is injective iff $\ker \varphi = \{e\}$.
- (b) Show that an ideal containing a unit is the whole ring. Conclude that the only ideals of a field are 0 and $\mathbb{K}$, and that every ring homomorphism from a field to a non-zero ring is injective.
- (c) Show that $\mathrm{ev}_i : \mathbb{R}[x] \rightarrow \mathbb{C}$, $p \mapsto p(i)$, is a surjective ring homomorphism with $\ker \mathrm{ev}_i = (x^2 + 1)$.
- (d) For which $m, n \geq 2$ is there a ring homomorphism $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ (with $1 \mapsto 1$)? Is there one $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$? One $\mathbb{Z}/12\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$?
- (e) Show that $\exp : (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$ is a group isomorphism, but that $(\mathbb{Q}, +)$ and $(\mathbb{Q}_{>0}, \cdot)$ are *not* isomorphic.

Exercise 1.12 (*) *Ring endomorphisms of $\mathbb{R}$.* Show that the only ring homomorphism $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is the identity. *Hint:* first show $\varphi|_{\mathbb{Q}} = \mathrm{id}$, then that φ maps squares to squares and is therefore monotone. Contrast this with $\mathbb{C}$: give a ring automorphism of $\mathbb{C}$ other than the identity.

Sheet 2: Linear Algebra

Exercise 2.1 (W) *Linear independence.*

- (a) Show that $v_1 = (1, 2, 3)$, $v_2 = (4, 5, 6)$, $v_3 = (7, 8, 9)$ are linearly dependent in $\mathbb{R}^3$. Find a basis of $\text{Span}(v_1, v_2, v_3)$ and extend it to a basis of $\mathbb{R}^3$.
- (b) In $\text{Maps}(\mathbb{R}, \mathbb{R})$, show that $1, \sin, \cos$ are linearly independent, while $1, \sin^2, \cos^2$ are linearly dependent.
- (c) Show that $1, \sqrt{2}, \sqrt{3}$ are linearly independent in $\mathbb{R}$ viewed as a $\mathbb{Q}$ -vector space.
- (d) (*) Show that for pairwise distinct $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ the functions $x \mapsto e^{\lambda_1 x}, \dots, e^{\lambda_n x}$ are linearly independent. Conclude that $\text{Maps}(\mathbb{R}, \mathbb{R})$ is infinite-dimensional.

Exercise 2.2 (W) *Computing dimensions.* Determine the dimension (with justification, e.g. by exhibiting a basis) of the following spaces:

- (a) $\mathbb{C}^n$ and $\text{Mat}(n \times n, \mathbb{C})$ as *real* vector spaces;
- (b) the real symmetric, the real skew-symmetric, and the real trace-free $n \times n$ matrices;
- (c) the Hermitian $n \times n$ matrices $\{A \in \text{Mat}(n \times n, \mathbb{C}) \mid \overline{A}^t = A\}$ as a real vector space. Is this a *complex* subspace of $\text{Mat}(n \times n, \mathbb{C})$?
- (d) $\mathbb{R}$ as a $\mathbb{Q}$ -vector space. *Hint:* countability.

Exercise 2.3 *Rank-nullity in practice.*

- (a) Let $\Phi : \mathbb{R}_3[x] \rightarrow \mathbb{R}^2$, $p \mapsto (p(1), p(-1))$. Find a basis of $\ker \Phi$, compute $\text{rank } \Phi$, and decide whether Φ is surjective.
- (b) Let $\Psi : \mathbb{R}_3[x] \rightarrow \mathbb{R}_3[x]$, $\Psi(p) = xp' - p$. Determine $\ker \Psi$ and $\text{im } \Psi$ and verify the dimension formula.
- (c) Let V be finite-dimensional and $\Phi \in \text{End}(V)$. Show that Φ is injective iff it is surjective iff it is bijective.
- (d) Show that (c) fails in infinite dimensions by considering $D = \frac{d}{dx}$ and $M_x : p \mapsto xp$ on $\mathbb{R}[x]$. Compute $D \circ M_x - M_x \circ D$.
- (e) Let $\dim V = n$ and $\Phi \in \text{End}(V)$ with $\Phi^2 = 0$. Show that $\text{rank } \Phi \leq n/2$.

Exercise 2.4 *Sums and intersections of subspaces.* Let $U, W \subseteq V$ be finite-dimensional subspaces.

- (a) Apply the dimension formula to $U \oplus W \rightarrow V$, $(u, w) \mapsto u + w$, to prove $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$.
- (b) Show that two distinct planes through the origin in $\mathbb{R}^3$ meet in a line, and that two 3-dimensional subspaces of $\mathbb{R}^5$ always intersect non-trivially.
- (c) Show that $V = U \oplus W$ (i.e. $U \oplus W \rightarrow V$ is an isomorphism) iff $U + W = V$ and $U \cap W = 0$.

Exercise 2.5 *Quotient spaces.*

- (a) Let $U = \text{Span}\{(1, 1, 1)\} \subset \mathbb{R}^3$. Find a linear map $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with $\ker \Phi = U$, conclude $\mathbb{R}^3/U \cong \mathbb{R}^2$, and give a basis of $\mathbb{R}^3/U$.
- (b) Use $D = \frac{d}{dx}$ to show $\mathbb{R}_n[x]/\mathbb{R} \cong \mathbb{R}_{n-1}[x]$, where $\mathbb{R}$ denotes the constants.

- (c) Show that $C^0([0,1])/\{f \mid f(0) = 0\} \cong \mathbb{R}$. (Note that this works although $C^0([0,1])$ is infinite-dimensional.)
- (d) (*) Let $U \subseteq W \subseteq V$ be subspaces. Use the universal property of the quotient to construct a surjection $V/U \rightarrow V/W$ and deduce $(V/U)/(W/U) \cong V/W$.

Exercise 2.6 *Direct sums and complements.*

- (a) Show $\text{Mat}(n \times n, \mathbb{R}) = \{\text{symmetric}\} \oplus \{\text{skew-symmetric}\}$ and $\text{Maps}(\mathbb{R}, \mathbb{R}) = \{\text{even}\} \oplus \{\text{odd}\}$.
- (b) Let $P \in \text{End}(V)$ with $P^2 = P$. Show that $V = \ker P \oplus \text{im } P$ and that P acts as the identity on $\text{im } P$.
- (c) Using Theorem 2.2, show that every subspace $U \subseteq V$ has a complement W (i.e. $V = U \oplus W$), and that any complement satisfies $W \cong V/U$. Compare with $2\mathbb{Z} \subset \mathbb{Z}$ (Exercise 1.7(d)).

Exercise 2.7 *Dual bases and annihilators.*

- (a) Compute the dual basis $(\alpha^1, \alpha^2, \alpha^3)$ of $v_1 = (1, 1, 0)$, $v_2 = (0, 1, 1)$, $v_3 = (1, 0, 1)$ in $\mathbb{R}^3$, writing each α^i as a row vector.
- (b) Show that $\text{ev}_0, \text{ev}_1, \text{ev}_2$ form a basis of $\mathbb{R}_2[x]^*$, and find the basis (p_0, p_1, p_2) of $\mathbb{R}_2[x]$ to which it is dual. What is the relation to polynomial interpolation?
- (c) For $U \subseteq V$ (with $\dim V = n$) define the annihilator $U^0 = \{\alpha \in V^* \mid \alpha|_U = 0\}$. Show that $\dim U^0 = n - \dim U$.
- (d) Let $\Phi : V \rightarrow W$ be linear between finite-dimensional spaces. Show that $\ker \Phi^* = (\text{im } \Phi)^0$, deduce $\text{rank } \Phi^* = \text{rank } \Phi$, and conclude (using Remark 2.8) that row rank equals column rank.
- (e) Show that $(\Phi \circ \Psi)^* = \Psi^* \circ \Phi^*$ and $\text{id}_V^* = \text{id}_{V^*}$.

Exercise 2.8 (*) *Duals in infinite dimensions.* Let $V = \mathbb{K}[x]$ and let $\alpha^k \in V^*$ be “the coefficient of x^k ”.

- (a) Show that $V^* \rightarrow \mathbb{K}^{\mathbb{N}_0}$, $\alpha \mapsto (\alpha(x^k))_{k \geq 0}$, is an isomorphism onto the space of *all* sequences.
- (b) Show that the α^k are linearly independent but do not span V^* . *Hint:* consider ev_1 .
- (c) For $\mathbb{K} = \mathbb{Q}$, show that V and V^* are not isomorphic. *Hint:* count elements.

Exercise 2.9 *Matrix representations.*

- (a) Let $\Phi : \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$, $\Phi(p)(x) = p(x+1)$. Find its matrix with respect to $(1, x, x^2)$, and use it to find the matrix of Φ^{-1} . What is Φ^{-1} ?
- (b) Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\text{ad}_A : \text{Mat}(2 \times 2, \mathbb{R}) \rightarrow \text{Mat}(2 \times 2, \mathbb{R})$, $X \mapsto AX - XA$. Find the matrix of ad_A with respect to $(E_{11}, E_{12}, E_{21}, E_{22})$, its rank, a basis of its kernel, and its trace.
- (c) For $D = \frac{d}{dx}$ on $\mathbb{K}_n[x]$ (Example 2.7), compute $\text{rank } D^k$ for all k and show $D^{n+1} = 0$.

Exercise 2.10 *Change of basis.*

- (a) Let $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $\Phi(x, y, z) = (x + y, y + z)$, and let $\tilde{B} = ((1, 0, 0), (1, 1, 0), (1, 1, 1))$, $\tilde{C} = ((1, 1), (1, -1))$. Compute the matrix $\tilde{T}$ of Φ with respect to $(\tilde{B}, \tilde{C})$ in two ways: directly, and via $\tilde{T} = U^{-1}TS$.

- (b) Show that similar matrices have the same trace, determinant and characteristic polynomial. Show that $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and I_2 have the same trace, determinant and characteristic polynomial, but are not similar.
- (c) Show that there are no $A, B \in \text{Mat}(n \times n, \mathbb{K})$ with $AB - BA = I_n$ if $\text{char } \mathbb{K} = 0$. Compare with Exercise 2.3(d).

Exercise 2.11 (W) *The dual map in coordinates.* For the map $\Phi : \mathbb{R}_2[x] \rightarrow \mathbb{R}^2$, $p \mapsto (p(1), p'(1))$ of Example 2.5, let (β^1, β^2) be the dual of the standard basis of $\mathbb{R}^2$ and $(\alpha^0, \alpha^1, \alpha^2)$ the dual of $(1, x, x^2)$. Express $\Phi^* \beta^1$ and $\Phi^* \beta^2$ in terms of the α^i and verify Remark 2.8.

Sheet 3: Bilinear and Hermitian Forms

Convention. Hermitian forms are antilinear in the first and linear in the second argument, as in the example $\langle v, w \rangle = \bar{v}^t w$ on $\mathbb{C}^n$. A Hermitian inner product is assumed positive definite.

Exercise 3.1 (W) *Gram matrices and change of basis.* Let β be the bilinear form on $\mathbb{R}^2$ with Gram matrix $B = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ with respect to the standard basis.

- Write out $\beta(x, y)$ explicitly. Is β symmetric? Non-degenerate? Positive definite?
- Compute the Gram matrix with respect to $\tilde{v}_1 = (1, 0)$, $\tilde{v}_2 = (-2, 1)$, both directly and via $T^t B T$. What is the signature of β ?
- Find all *isotropic* vectors, i.e. all v with $\beta(v, v) = 0$.
- Compute the pullback $\varphi^* \beta$ along $\varphi : \mathbb{R} \rightarrow \mathbb{R}^2$, $t \mapsto (t, t)$.

Exercise 3.2 *Symmetric, skew-symmetric and alternating forms.* Let V be a finite-dimensional $\mathbb{K}$ -vector space.

- If $\text{char } \mathbb{K} \neq 2$, show that every bilinear form is uniquely the sum of a symmetric and a skew-symmetric form.
- Show that every alternating bilinear form is skew-symmetric ($\beta(v, w) = -\beta(w, v)$), and that the converse holds if $\text{char } \mathbb{K} \neq 2$.
- Give a skew-symmetric form on $\mathbb{F}_2$ that is not alternating.
- Let $\text{char } \mathbb{K} \neq 2$. Show that a skew-symmetric matrix of odd size has determinant 0. Conclude that a symplectic vector space (over such $\mathbb{K}$) has even dimension.

Exercise 3.3 *Sylvester's law of inertia in practice.*

- Let $q(x, y, z) = x^2 + y^2 + z^2 + 2xy + 2yz$ be the quadratic form of a symmetric bilinear form β on $\mathbb{R}^3$. Write down the Gram matrix, determine the signature by completing squares, and find a basis in which the Gram matrix is $\text{diag}(1, 1, -1)$.
- For which $a \in \mathbb{R}$ is the form with Gram matrix $\begin{pmatrix} 1 & a & 0 \\ a & 1 & a \\ 0 & a & 1 \end{pmatrix}$ positive definite?
- Let η be the Lorentz inner product on $\mathbb{R}^4$. Find a non-zero v with $\eta(v, v) = 0$. Show that there is no 2-dimensional subspace on which η is positive definite. *Hint:* mimic Part 2 of the proof of Theorem 3.3.

Exercise 3.4 *The trace form.* On $V = \text{Mat}(n \times n, \mathbb{R})$ consider $\beta(A, B) = \text{tr}(AB)$ and $\langle A, B \rangle = \text{tr}(A^t B)$.

- Show that $\langle \cdot, \cdot \rangle$ is an inner product and that (E_{ij}) is an orthonormal basis.
- Show that β is symmetric and non-degenerate. *Hint:* compute $\beta(A, E_{ji})$.
- Show that symmetric and skew-symmetric matrices are β -orthogonal, and determine the signature of β .

Exercise 3.5 *Gram-Schmidt and best approximation.* Equip $\mathbb{R}_2[x]$ with $\langle p, q \rangle = \int_{-1}^1 p(t)q(t) dt$.

- Apply the Gram-Schmidt procedure to $(1, x, x^2)$ to obtain an orthonormal basis.
- Find the polynomial p of degree ≤ 1 minimising $\int_{-1}^1 (t^2 - p(t))^2 dt$. *Hint:* orthogonal projection, and $\lambda_i = \langle v_i, v \rangle$ for an ONB.

Exercise 3.6 *Cauchy–Schwarz and the triangle inequality.* Let $(V, \langle \cdot, \cdot \rangle)$ be a real or complex inner product space and $\|v\| = \sqrt{\langle v, v \rangle}$.

- (a) Prove the Cauchy–Schwarz inequality in the real case by considering $\|v + tw\|^2$ for $t \in \mathbb{R}$, and show that equality holds iff v, w are linearly dependent. Adapt the proof to the complex case.
- (b) Deduce the triangle inequality $\|v + w\| \leq \|v\| + \|w\|$.
- (c) Show that $(a_1 + \cdots + a_n)^2 \leq n(a_1^2 + \cdots + a_n^2)$ and $(\int_0^1 f)^2 \leq \int_0^1 f^2$ for continuous f .

Exercise 3.7 *Polarisation.*

- (a) For a symmetric bilinear form β on a real vector space with $q(v) = \beta(v, v)$, show $\beta(v, w) = \frac{1}{2}(q(v + w) - q(v) - q(w))$.
- (b) For a sesquilinear form h on a complex vector space (not necessarily Hermitian), with $q(v) = h(v, v)$, show that $h(v, w) = \frac{1}{4} \sum_{k=0}^3 i^{-k} q(v + i^k w)$.
- (c) Deduce: on a complex inner product space, if $\langle v, \Phi v \rangle = 0$ for all v , then $\Phi = 0$. Show by an example that this fails over $\mathbb{R}$.
- (d) Deduce: on a complex inner product space, Φ is self-adjoint iff $\langle v, \Phi v \rangle \in \mathbb{R}$ for all v .
- (e) Show that a linear map between real inner product spaces that preserves norms is an isometry.

Exercise 3.8 *Adjoint.* Let V, W be finite-dimensional inner product spaces over $\mathbb{R}$ or $\mathbb{C}$.

- (a) Show that the adjoint $\Phi^\dagger$ is unique, and that $(\Phi\Psi)^\dagger = \Psi^\dagger\Phi^\dagger$, $(\lambda\Phi)^\dagger = \bar{\lambda}\Phi^\dagger$ and $(\Phi^\dagger)^\dagger = \Phi$.
- (b) If G is the Gram matrix of the inner product with respect to a (not necessarily orthonormal) basis of V and T represents $\Phi \in \text{End}(V)$, show that $\Phi^\dagger$ is represented by $G^{-1}\bar{T}^t G$.
- (c) Let $V = \mathbb{R}_1[x]$ with $\langle p, q \rangle = \int_0^1 pq \, dt$ and $D = \frac{d}{dx}$. Compute $D^\dagger$ using (b) with the basis $(1, x)$. Check your answer, and explain why $D^\dagger \neq -D$ despite “integration by parts”.
- (d) Show that $\ker \Phi^\dagger = (\text{im } \Phi)^\perp$, and deduce the orthogonal decomposition $V = \ker \Phi \oplus \text{im } \Phi^\dagger$ for $\Phi : V \rightarrow W$.

Exercise 3.9 *Hermitian 2×2 matrices and the Pauli matrices.* Let $\mathcal{H}$ be the real vector space of Hermitian 2×2 matrices and $E_0, \dots, E_3$ as in the lecture.

- (a) Show that $E_0, \dots, E_3$ form a basis of $\mathcal{H}$.
- (b) For $X = x_0 E_0 + x_1 E_1 + x_2 E_2 + x_3 E_3$, compute $\text{tr } X$ and $\det X$. Which bilinear form from the lecture appears?
- (c) Show that $E_j^2 = I$ for $j = 1, 2, 3$ and $E_1 E_2 = i E_3$. Determine the eigenvalues of E_1, E_2, E_3 , and show that each E_j is both Hermitian and unitary.
- (d) Let $U \in \text{Mat}(2 \times 2, \mathbb{C})$ be unitary. Show that $X \mapsto UXU^\dagger$ is a linear map $\mathcal{H} \rightarrow \mathcal{H}$ that preserves $\det$. Interpret this using (b).

Exercise 3.10 *Classes of operators.* Let V be a finite-dimensional complex inner product space.

- (a) Show that self-adjoint, skew-adjoint and unitary operators are normal. Give a normal matrix that is none of these three, and a matrix that is not normal.

- (b) Show that the eigenvalues of a skew-adjoint operator are purely imaginary, and that $|\det U| = 1$ for unitary U .
- (c) Show that every $A \in O(2)$ is either a rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ or a reflection $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$.
- (d) (*) (Cayley transform.) Let A be skew-adjoint. Show that $I \pm A$ are invertible and that $U = (I + A)(I - A)^{-1}$ is unitary. *Hint:* compute $\|(I \pm A)v\|^2$.

Exercise 3.11 *Symplectic forms and complex structures.*

- (a) On $\mathbb{C}^n$ with the standard Hermitian inner product, let $g = \operatorname{Re}\langle \cdot, \cdot \rangle$ and $\omega = -\operatorname{Im}\langle \cdot, \cdot \rangle$. Identifying $\mathbb{C}^n \cong \mathbb{R}^{2n}$ via $z_k = x_k + iy_k$, show that g is the Euclidean inner product, that $\omega(v, w) = g(v, iw)$, and that ω is a symplectic form.
- (b) Let W be a real vector space of dimension m and $V = W \oplus W^*$ with $\langle \cdot, \cdot \rangle$ and ω as in the lecture. Show that ω is symplectic. Using a basis (w_i) of W with dual basis (α^i) , show that the vectors $(w_i, \pm \alpha^i)$ are pairwise orthogonal, and conclude that $\langle \cdot, \cdot \rangle$ has signature (m, m) .
- (c) Show that $W \oplus 0$ and $0 \oplus W^*$ are subspaces of half dimension on which ω vanishes (Lagrangian subspaces).

Sheet 4: Determinants, Eigenvalues and Canonical Forms

Exercise 4.1 (W) *Determinants from the axioms.*

- Using only D1–D3, show that swapping two rows changes the sign of $\det$, that a matrix with a zero row has determinant 0, and derive D6.
- Show that the determinant of a triangular matrix is the product of its diagonal entries.
- Compute $\det \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}$. More generally, let D_n be the determinant of the $n \times n$ tridiagonal matrix with 2 on the diagonal and 1 on the off-diagonals. Find a recursion for D_n and solve it.
- Show that $\det \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{pmatrix} = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)$. Deduce that through three points with distinct x -coordinates there is exactly one polynomial of degree ≤ 2 .
- (*) Prove the general Vandermonde formula $\det(x_i^{j-1})_{i,j=1}^n = \prod_{i < j} (x_j - x_i)$.

Exercise 4.2 *The Leibniz formula and permutation matrices.*

- For $n = 3$, list the six permutations with their signs and recover Sarrus' rule.
- Use the Leibniz formula to prove $\det A^t = \det A$ (D8).
- For $\sigma \in S_n$ let P_σ be the matrix with $P_\sigma e_i = e_{\sigma(i)}$. Show that $P_\sigma P_\tau = P_{\sigma\tau}$ and $\det P_\sigma = \text{sgn } \sigma$. Conclude that $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a group homomorphism.

Exercise 4.3 *Diagonalisation and matrix powers.*

- Diagonalise $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$, i.e. find S invertible and D diagonal with $A = SDS^{-1}$, and compute A^n for all $n \in \mathbb{N}$.
- (Fibonacci numbers.) Let $F_0 = 0$, $F_1 = 1$, $F_{n+1} = F_n + F_{n-1}$. Find a matrix M with $\begin{pmatrix} F_{n+1} \\ F_n \end{pmatrix} = M \begin{pmatrix} F_n \\ F_{n-1} \end{pmatrix}$, diagonalise it, and derive Binet's formula $F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$ with $\varphi, \psi = \frac{1 \pm \sqrt{5}}{2}$.

Exercise 4.4 *When is a matrix diagonalisable?*

- For which $a, b \in \mathbb{R}$ is $M = \begin{pmatrix} 1 & a & b \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ diagonalisable?
- Let R_θ be the rotation matrix with $\theta \notin \pi\mathbb{Z}$. Show that R_θ has no real eigenvalues. The condition “algebraic = geometric multiplicity for all eigenvalues” of Theorem 4.1 then holds vacuously, yet R_θ is not diagonalisable over $\mathbb{R}$. Which hypothesis is implicitly needed in Theorem 4.1? Is R_θ diagonalisable over $\mathbb{C}$?
- Let $\Phi \in \text{End}(V)$, $\dim V < \infty$. Show: if $\Phi^2 = \Phi$, then Φ is diagonalisable with eigenvalues in $\{0, 1\}$. If $\Phi^2 = \text{id}$ and $\text{char } \mathbb{K} \neq 2$, then Φ is diagonalisable with eigenvalues in $\{\pm 1\}$. If $\Phi^2 = 0$, then Φ is diagonalisable iff $\Phi = 0$.

Exercise 4.5 *Eigenvalues and multiplicities: theory.* Let $\dim V = n < \infty$ and $\Phi \in \text{End}(V)$.

- Show that eigenvectors for pairwise distinct eigenvalues are linearly independent.
- Show that $1 \leq g_\lambda \leq a_\lambda$. *Hint:* extend a basis of E_λ and use D7.

- (c) Over $\mathbb{C}$, show that $\det \Phi = \prod_{\lambda} \lambda^{a_{\lambda}}$ and $\text{Tr } \Phi = \sum_{\lambda} a_{\lambda} \lambda$. *Hint:* compare two expressions for the coefficients of χ_{Φ} .
- (d) Show that AB and BA have the same non-zero eigenvalues.
- (e) (*) Show that AB and BA even have the same characteristic polynomial. *Hint:* conjugate $\begin{pmatrix} AB & 0 \\ B & 0 \end{pmatrix}$ by $\begin{pmatrix} I & A \\ 0 & I \end{pmatrix}$.

Exercise 4.6 Nilpotent endomorphisms. Let $\dim V = n$ and $N \in \text{End}(V)$ nilpotent, i.e. $N^k = 0$ for some k .

- (a) Show that 0 is the only eigenvalue of N , and that N is diagonalisable only if $N = 0$.
- (b) Show that $I + N$ is invertible, and give its inverse.
- (c) Show that if $\ker N^j = \ker N^{j+1}$ for some j , then $\ker N^j = \ker N^{j+i}$ for all $i \geq 0$. Deduce that $N^n = 0$.

Exercise 4.7 The spectral theorem in practice.

- (a) Find an orthogonal matrix Q such that $Q^t A Q$ is diagonal, for $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$.
- (b) Find a unitary U such that $U^{\dagger} A U$ is diagonal, for $A = \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}$.
- (c) Diagonalise the rotation R_{θ} over $\mathbb{C}$ by a unitary matrix.
- (d) Prove the converses: a unitarily diagonalisable complex matrix is normal, and an orthogonally diagonalisable real matrix is symmetric.
- (e) Show directly that eigenvectors of a self-adjoint operator for distinct eigenvalues are orthogonal.

Exercise 4.8 Consequences of the spectral theorem. Let $A \in \text{Mat}(n \times n, \mathbb{R})$ be symmetric and $\beta_A(x, y) = x^t A y$.

- (a) Show that the signature of β_A is (number of positive eigenvalues, number of negative eigenvalues), counted with multiplicity. In particular, β_A is positive definite iff all eigenvalues of A are positive. Use this to recheck Exercise 3.3(a).
- (b) Show that $\max_{\|x\|=1} x^t A x$ is the largest eigenvalue of A .
- (c) Show that if A is positive definite, there is a positive definite B with $B^2 = A$. (*) Show that B is unique.
- (d) (*) Let A, B be commuting self-adjoint operators on a complex inner product space. Show that they have a common orthonormal basis of eigenvectors.

Exercise 4.9 Jordan canonical form.

- (a) Let $A = \begin{pmatrix} 3 & 1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$. Determine the characteristic polynomial, the algebraic and geometric multiplicities, the Jordan canonical form J , and a basis (v_1, v_2, v_3) in which A is represented by J . Find the decomposition $A = D + N$ of Theorem 4.3.
- (b) List all possible Jordan forms of a 4×4 matrix with $\chi_A(t) = (t-1)^4$. Show that $\dim \ker(A-I)$ alone does not determine the Jordan form, but that $\dim \ker(A-I)$ together with $\dim \ker(A-I)^2$ does.
- (c) Compute $J_{\lambda, n}^k$ using $J_{\lambda, n} = \lambda I + N$ and the binomial theorem. Write down $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)^k$ explicitly.
- (d) (*) Prove the Cayley–Hamilton theorem over $\mathbb{C}$, i.e. $\chi_A(A) = 0$, using the Jordan form.
- (e) (*) Show that every $A \in \text{Mat}(n \times n, \mathbb{C})$ is similar to A^t .

Sheet 5: Tensor Products and Exterior Algebra

Exercise 5.1 (W) *Pure and entangled tensors.*

- Show that $v \otimes 0 = 0$ and $(\lambda v) \otimes w = v \otimes (\lambda w)$ in $V \otimes W$.
- Show that $\mathbb{K}^m \otimes \mathbb{K}^n \cong \text{Mat}(m \times n, \mathbb{K})$ via $v \otimes w \mapsto vw^t$, and that under this isomorphism the pure tensors $v \otimes w$ correspond exactly to the matrices of rank ≤ 1 .
- Show that $e_1 \otimes e_1 + e_2 \otimes e_2 \in \mathbb{K}^2 \otimes \mathbb{K}^2$ is not a pure tensor. More generally, show that $\sum_{i,j} a_{ij} e_i \otimes e_j$ is pure iff $\det(a_{ij}) = 0$.
- The *tensor rank* of $t \in V \otimes W$ is the minimal r such that t is a sum of r pure tensors. Show that for $V = \mathbb{K}^m$, $W = \mathbb{K}^n$ it equals the rank of the corresponding matrix.
- In quantum mechanics a state in $\mathbb{C}^2 \otimes \mathbb{C}^2$ is *entangled* if it is not a pure tensor. Show that $\frac{1}{\sqrt{2}}(e_1 \otimes e_2 - e_2 \otimes e_1)$ is entangled.

Exercise 5.2 *Using the universal property.* Let V, W be finite-dimensional.

- Show that $\mathbb{K} \otimes V \cong V$ and $V \otimes W \cong W \otimes V$ by constructing mutually inverse linear maps via the universal property.
- Construct a linear map $c : V^* \otimes V \rightarrow \mathbb{K}$ with $c(\alpha \otimes v) = \alpha(v)$ (“contraction”). Show that under the isomorphism $V^* \otimes V \cong \text{End}(V)$ of the lecture, c corresponds to the trace. (This gives a basis-free definition of the trace.)
- Show that under $V^* \otimes V \cong \text{End}(V)$ the identity corresponds to $\sum_i \alpha^i \otimes v_i$ for any basis (v_i) with dual basis (α^i) . Conclude that this element does not depend on the choice of basis.

Exercise 5.3 *Tensor products of linear maps: the Kronecker product.* Let $A \in \text{Mat}(n \times n, \mathbb{K})$ represent $\Phi : V \rightarrow V$ and $B \in \text{Mat}(m \times m, \mathbb{K})$ represent $\Psi : W \rightarrow W$, with respect to bases (v_j) and (w_l) .

- Explain why $\Phi \otimes \Psi$ is well defined, and show $(\Phi \otimes \Psi) \circ (\Phi' \otimes \Psi') = \Phi\Phi' \otimes \Psi\Psi'$.
- Show that with respect to $(v_1 \otimes w_1, \dots, v_1 \otimes w_m, v_2 \otimes w_1, \dots, v_n \otimes w_m)$, the map $\Phi \otimes \Psi$ is represented by the block matrix $A \otimes B = (a_{ij}B)_{i,j}$. Compute $A \otimes B$ for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- Show that $\text{Tr}(\Phi \otimes \Psi) = \text{Tr } \Phi \cdot \text{Tr } \Psi$, and that if $\Phi v = \lambda v$ and $\Psi w = \mu w$ then $(\Phi \otimes \Psi)(v \otimes w) = \lambda \mu v \otimes w$.
- (*) Show that $\det(A \otimes B) = (\det A)^m (\det B)^n$. Check this in the example from (b).

Exercise 5.4 *Tensor products of function spaces.* Let X, Y be finite sets.

- Prove that $\text{Maps}(X, \mathbb{K}) \otimes \text{Maps}(Y, \mathbb{K}) \cong \text{Maps}(X \times Y, \mathbb{K})$ via $(f \otimes g)(x, y) = f(x)g(y)$. *Hint:* use the bases (δ_x) , (δ_y) .
- Let $X = Y = \{0, 1, 2\} \subset \mathbb{R}$. Determine the tensor rank of $h(x, y) = x + y$ and of $k(x, y) = \delta_{xy}$, viewed as elements of $\text{Maps}(X, \mathbb{R}) \otimes \text{Maps}(Y, \mathbb{R})$.

Exercise 5.5 *Computing with the wedge product.* Let $(e_1, \dots, e_n)$ be a basis of V with dual basis $(x^1, \dots, x^n)$.

- In $\Lambda^3(\mathbb{R}^3)^*$, express $(x^1 + 2x^2) \wedge (x^2 - x^3) \wedge (x^1 + x^3)$ as a multiple of $x^1 \wedge x^2 \wedge x^3$.

- (b) Show that $\alpha \wedge \alpha = 0$ for $\alpha \in V^*$. Let $\omega = x^1 \wedge x^2 + x^3 \wedge x^4 \in \Lambda^2(\mathbb{R}^4)^*$. Compute $\omega \wedge \omega$. Is ω of the form $\alpha \wedge \beta$ with $\alpha, \beta \in V^*$?
- (c) Evaluate $(x^1 \wedge x^2)(v, w)$ for $v, w \in \mathbb{R}^2$ and interpret the result geometrically.
- (d) Show that $\alpha^1, \dots, \alpha^k \in V^*$ are linearly independent iff $\alpha^1 \wedge \dots \wedge \alpha^k \neq 0$.
- (e) Verify Remark 4.6 directly for $\omega = x^1$, $\eta = x^2$ and for $\omega = x^1$, $\eta = x^2 \wedge x^3$.

Exercise 5.6 *Dimensions and symmetric versus alternating tensors.*

- (a) Show that $\sum_{k=0}^n \dim \Lambda^k V^* = 2^n$ and that $\Lambda^k V^* = 0$ for $k > n$.
- (b) Let $\text{char } \mathbb{K} \neq 2$. In $V \otimes V$ let $S^2 = \text{Span}\{v \otimes w + w \otimes v\}$ and $A^2 = \text{Span}\{v \otimes w - w \otimes v\}$. Show that $V \otimes V = S^2 \oplus A^2$, and compute both dimensions. Relate this to Exercise 2.6(a).

Exercise 5.7 (*) *The determinant via the top exterior power.* Let $\dim V = n$ and $\Phi \in \text{End}(V)$. Define $\Phi^* : \Lambda^n V^* \rightarrow \Lambda^n V^*$ by $(\Phi^* \omega)(v_1, \dots, v_n) = \omega(\Phi v_1, \dots, \Phi v_n)$.

- (a) Show that $\Phi^* \omega = \det(\Phi) \omega$ for all $\omega \in \Lambda^n V^*$.
- (b) Deduce $\det(\Phi \Psi) = \det \Phi \det \Psi$ without using any matrix computation.

Exercise 5.8 *Inner products on Λ^2 and the Hodge star on $\mathbb{R}^3$.* Let (e_1, e_2, e_3) be the standard orthonormal basis of $\mathbb{R}^3$ and $\text{vol} = e_1 \wedge e_2 \wedge e_3$.

- (a) Show that $\|v \wedge w\|^2 = \|v\|^2 \|w\|^2 - \langle v, w \rangle^2$ for $v, w \in \mathbb{R}^n$. Interpret the result.
- (b) Compute $*$ on all basis elements $1, e_i, e_i \wedge e_j$, vol of $\Lambda^\bullet \mathbb{R}^3$, and check that $** = \text{id}$.
- (c) Verify $\alpha \wedge * \beta = \langle \alpha, \beta \rangle \text{vol}$ for $\alpha = \beta = e_1 \wedge e_2$ and for $\alpha = e_1 \wedge e_2$, $\beta = e_1 \wedge e_3$.
- (d) Show that $*(v \wedge w) = v \times w$ (the cross product). Using (a) and the fact that $*$ is an isometry, deduce Lagrange's identity $\|v \times w\|^2 = \|v\|^2 \|w\|^2 - \langle v, w \rangle^2$.

Exercise 5.9 *The Hodge star in dimensions 2 and 4.*

- (a) On $\mathbb{R}^2$ with $\text{vol} = e_1 \wedge e_2$, compute $*$ on $\Lambda^1 \mathbb{R}^2$ and show that $** = -\text{id}$ there. Which familiar operation on $\mathbb{C} \cong \mathbb{R}^2$ is $*$?
- (b) On $\mathbb{R}^4$ with $\text{vol} = e_1 \wedge e_2 \wedge e_3 \wedge e_4$, compute $*$ on the basis of $\Lambda^2 \mathbb{R}^4$ and show that $*^2 = \text{id}$ on $\Lambda^2 \mathbb{R}^4$. Find bases of the eigenspaces $\Lambda_{\pm}^2$ of $*$ (self-dual and anti-self-dual 2-forms) and their dimensions.
- (c) (*) Show that on $\Lambda^k \mathbb{R}^n$ (Euclidean, orthonormal basis) $** = (-1)^{k(n-k)} \text{id}$.